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Website: <https://appliedcryptography.page>

Quiz 1.5: Chosen-Plaintext & Chosen-Ciphertext Attacks

This completely optional quick quiz acts as a learning aid to help you find out how much you absorbed from our Chosen-Plaintext & Chosen-Ciphertext Attacks session.^a Remember, this is just for fun and to help you identify areas you might want to review. Each question has three possible answers, but only one is correct. Take your time, and when you're done, check your answers against the answer key at the end. Good luck!

^a<https://appliedcryptography.page/slides/#1-5>

Questions

1. What does CPA security guarantee about an encryption scheme?
 - A. Even if attackers can encrypt chosen messages, ciphertexts appear random
 - B. The ciphertext is always shorter than the plaintext
 - C. The encryption key can be recovered from the ciphertext
2. What information is allowed to leak even in a CPA-secure encryption scheme?
 - A. The length of the message
 - B. The first byte of the plaintext
 - C. The encryption key
3. Why does deterministic encryption always fail CPA security?
 - A. The key is too short
 - B. It uses weak encryption algorithms
 - C. The same message always encrypts to the same ciphertext
4. What is the "Golden Rule of PRFs" in cryptographic constructions?
 - A. Security depends on ensuring distinct inputs to the PRF
 - B. PRFs should only be used for authentication
 - C. Always use the longest possible key
5. What makes CTR mode different from CBC mode?
 - A. CTR mode is sequential and cannot be parallelized
 - B. CTR mode is highly parallelizable and acts like a stream cipher
 - C. CTR mode requires padding
6. In a format-oracle attack, what does the oracle reveal?
 - A. A small piece of information about the decrypted plaintext's format
 - B. The encryption key
 - C. The complete plaintext
7. What property makes CTR mode vulnerable to chosen-ciphertext attacks?

- A. It uses weak encryption
 - B. It requires too much memory
 - C. It is malleable - bit flips in ciphertext cause predictable changes in plaintext
8. In the null-oracle attack, how many queries are needed to recover one byte?
- A. Exactly 256 queries
 - B. At most 255 queries
 - C. At least 1000 queries
9. What is the difference between CPA and CCA security?
- A. They are the same thing
 - B. CPA is stronger than CCA
 - C. CCA additionally protects against adversaries who can decrypt chosen ciphertexts
10. What does a Message Authentication Code (MAC) provide?
- A. Only compression
 - B. Integrity and authenticity
 - C. Only confidentiality
11. Which MAC-and-encrypt combination is always CCA-secure?
- A. Encrypt-and-MAC
 - B. MAC-then-encrypt
 - C. Encrypt-then-MAC
12. Why is Encrypt-and-MAC not even CPA-secure?
- A. The MAC key is exposed
 - B. The encryption is too weak
 - C. The MAC is computed on plaintext, leaking equality information
13. What is Authenticated Encryption (AE)?
- A. Encryption that provides both confidentiality and authenticity
 - B. Encryption that requires user authentication
 - C. Encryption that uses public key certificates
14. What is the most widely used AEAD scheme in modern protocols?
- A. Triple DES
 - B. AES-CBC with HMAC
 - C. AES-GCM
15. What happens if you reuse a nonce in AES-GCM?
- A. Complete loss of both confidentiality and authentication
 - B. Nothing, it's still secure
 - C. Only confidentiality is lost
16. What is the purpose of associated data in AEAD?
- A. To store the encryption key
 - B. To compress the ciphertext
 - C. To bind context information without encrypting it
17. How does ChaCha20-Poly1305 compare to AES-GCM on devices without AES hardware?
- A. They perform identically
 - B. It's much slower

- C. It has excellent performance
18. What type of attack can AEAD schemes still be vulnerable to?
- A. Replay attacks
 - B. Brute force attacks on the ciphertext
 - C. Frequency analysis
19. What does the Fujisaki-Okamoto transform do?
- A. Increases the key size
 - B. Converts symmetric encryption to asymmetric
 - C. Converts CPA-secure public-key encryption to CCA-secure
20. What is key commitment in authenticated encryption?
- A. Publicly revealing part of the key
 - B. A ciphertext should only decrypt correctly under the key that created it
 - C. Committing to use the same key forever

Answers

Question	Key
1	A
2	A
3	C
4	A
5	B
6	A
7	C
8	B
9	C
10	B
11	C
12	C
13	A
14	C
15	A
16	C
17	C
18	A
19	C
20	B